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Título/Title

Program AIDA User's Manual
Version 4

Palabras clave/Key words **MANUALS / CROSS SECTIONS / COMPUTER PROGRAMS**	Clasificación acceso Access class **P1** Registro de Revisiones Revision Record **Pag. 2**

Resumen/Summary

The purpose of this Technical Report is to define instruction for use the AIDA program.

This Program calculates the geometric properties (area, position of the centre of gravity, moments of inertia and principal axes) of flat sections with any shape.

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1.- Technical sheet of the program

Technical Note **NT-T-ADP-09007**

PROGRAM	REFERENCE	DATE
AIDA		2009

OBJECTIVE

This program calculates the geometric properties (area, position of the centre of gravity, moments of inertia and principal axes) of flat sections with any shape.

LANGUAGE	COMPUTER
Fortran 77	Sun Solaris
NUMBER OF LINES (APROX.)	EXECUTION MODE
1600	Interactive
AUTHOR	VERSION
Darío Fernández Villalba	4.0

ADDITIONAL DOCUMENTS

Technical Note: NT–T–ADS–95015.
Theoretical Manual of the AIDA Program. Version 1.0.

2.- Introduction

This Manual describes the use of the AIDA program, version 4 and includes several examples for clarification.

This program calculates the geometrical properties (area, position of the centre of gravity, moments of inertia and principal axes of inertia) for flat sections.

The section to calculate can have any shape (closed or open, thin or thick wall) but the shape must be exclusively composed by straight lines and/or segments of a circle. Any other shape can be approximated by these types of lines.

Differences between versions:

- Version 1.0:

  Initial version developed in Fortran 77 for Digital VAX computer.

- Version 2.0:

  The program is adapted to the new Sun Solaris computer.

- Version 3.0:

  The program is re-arranged and only one section can be calculated in each run.

  Now, the user can choose the language for screen dialogs and output files, between Spanish and English.

  The program source is restructured to allow the creation of a Dynamic Link Library DLL for Windows operating system.

  In parallel with this, a Windows application has been developed as Graphic User Interface GUI for the program FATIGA. This application uses the same input files and provides the same output results. The fatigue analysis is performed by calling to the DLL, compiled from the same source as the official Fortran version.

- Version 4.0:

  The program is re-arranged. The format of the input file is different from previous versions of the program.

  The output results are now provided in a single text file.

  The GUI has been improved. The output results are provided on the screen. It is also possible to write them in a text file, identical to the output from the official program.

  The GUI allows to the user the obtainment of the section shape in a graphic file and the output results, including a drawing of the section, in the printer.

3.- Execution of the program

Prior to the program execution, it is necessary to create the input data file, as described in section 4. The program can be then executed by typing its **full path name**.

If the software symbol table has been previously assigned (consult with the System Manager in order to know how to assign this symbol table), it can be executed by typing the command:

```
$aida
```

The program asks for the language to use and the name of the input and output files, as shown in the following example:

```
Idioma / Language:
==================

  1 ... ESPAÑOL

  2 ... ENGLISH

Idioma / Language: 2

Name of the input file:  Example.dat

Name of the output file:  Example.txt

Units:
======

  1 ... mm

  2 ... cm

  3 ... in

Units: 1

Reading input data file
Calculating section properties
Writing the output file

AIDA 4.0. (c) Airbus Military 2009
```

The CPU time for calculation is negligible.

It is possible to execute the program with a single command, including the input and output filenames, the units and the language as parameters. The above example can be executed as follows:

```
$aida Example.dat Example.txt mm 2
```

4.- Input data file

This file contains all the necessary data that defines the section. It can be defined in two different ways:

- <u>Section defined by segments:</u> In this case, the structure of this group is the following:

Line 1 ----------> 1 "Optional commentary"

[Ns] lines ------->	Ns	Xi(1)	Yi(1)	Xf(i)	Yf(i)	R(1)	t(i)
		Xi(2)	Yi(2)	Xf(2)	Yf(2)	R(2)	t(2)
		……	……	……	……	……	……
		……	……	……	……	……	……
		Xi(Ns)	Yi(Ns)	Xf(Ns)	Yf(Ns)	R(Ns)	t(Ns)

Where: Ns = Number of segments to define the section.

Xi(i),Yi(i) = X and Y coordinates of the initial end of the segment.

Xf(i),Yf(i) = X and Y coordinates of the final end of the segment.

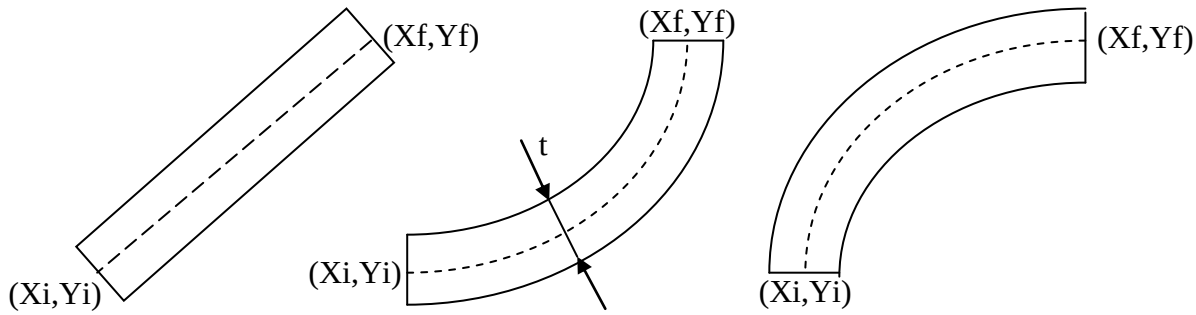
R(i) = Radius of the mid line of the segment, with the following conventions:

R = 0: Straight segment.

R < 0: Circular arc with radius R in which the arc from the start point to the end point is in the clockwise direction.

R > 0: Circular arc with radius R in which the arc from the start point to the end point is in the anticlockwise direction.

t(i) = Thickness of the segment.



<u>R = 0</u> <u>R > 0</u> <u>R < 0</u>

For circular arc segments, initial and final points must be located in such a way that the arc is no larger than 180º. If it is longer, it is necessary to divide the segment, establishing intermediate points.

The segments do not need to be specified in any particular order.

- <u>Section defined by contour lines:</u> In this case, the structure of this group is the following:

Line 1 ----------> 2 "Optional commentary"

[Ns] lines ------->	Np	X(1)	Y(1)	R(1)
		X(2)	Y(2)	R(2)
		……	……	……
		……	……	……
		X(Np)	Y(Np)	R(Np)

Where: Np = Number of points to define the contour of the section.

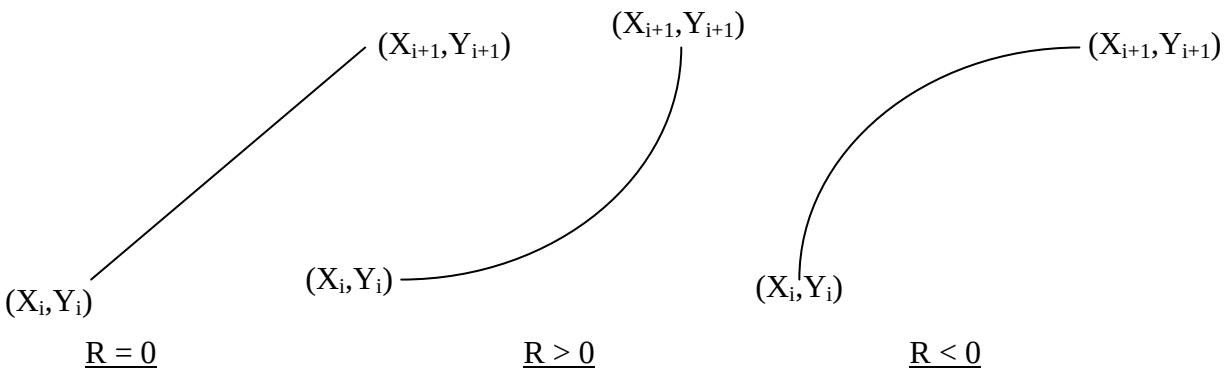
X(i),Y(i) = X and Y coordinates of the point.

R(i) = Radius of the arc joining the point "i" and the point "i+1", with the following conventions:

R = 0: Straight line.

R < 0: Circular arc with radius R in which the arc from the point "i" to the point "i+1" is in the clockwise direction.

R > 0: Circular arc with radius R in which the arc from the point "i" to the point "i+1" is in the anticlockwise direction.



<u>R = 0</u> <u>R > 0</u> <u>R < 0</u>

For circular arc lines, the end points must be defined so that the arc is no larger than 180º. Otherwise the arc must be split into subsections by selecting intermediate points.

In this case the contour lines must be in the correct order, since the program "builds up" the section by joining each point to the start point of the next segment according to the order in which they were provided to the program.

5.- Output results file

This file contains the following information:

- Input data:
  - Method to define the geometry.
  - Coordinates and parameters defining the geometry of the section.
- Section properties:
  - Area of the section.
  - Centre of gravity position.
  - Moments of inertia in the centre of gravity.
  - Moments of inertia in principal axes.
  - Angle between reference axes and principal axes.

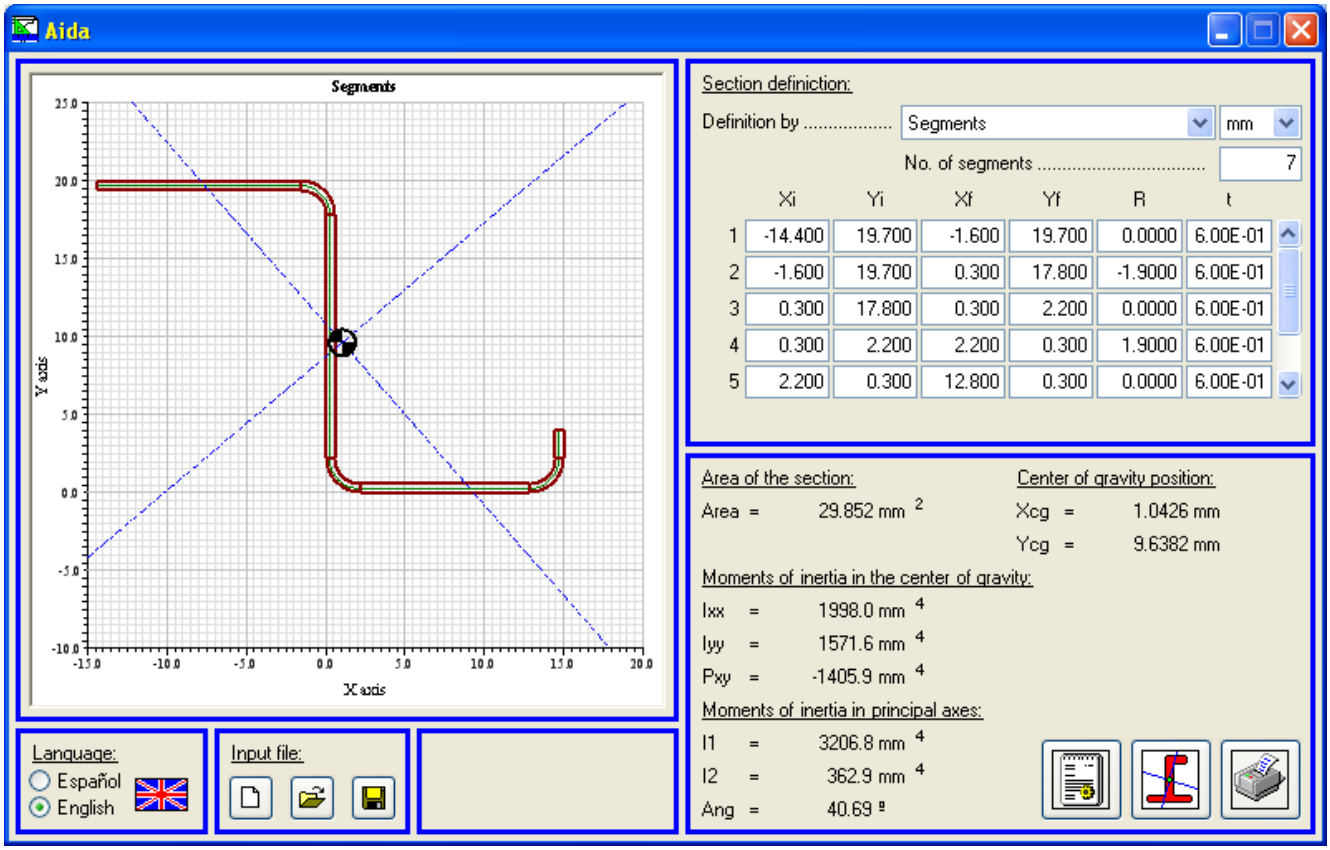
6.- Windows application

An application has been developed in Visual Basic for Windows environment. The original routines of the program for input data, calculation and output results have been included in a Dynamic Link Library, which is used by this application. Therefore the same results will be obtained, with only differences due to rounding.

Since this is a Windows application, it must be installed in the user's PC before it can be executed. For installation, please contact with the system manager.

The application can be executed by "double click" in the file **Aida.exe**, located in the PC's server, or by means a **link** in the "desktop" or in the "start menu".

The input data must be typed in the application form or retrieved from an input file. The form, after loading the data corresponding to the example 1 (see section 7), has the following appearance:



The program automatically recalculates the section properties at the time that some input data are changed.

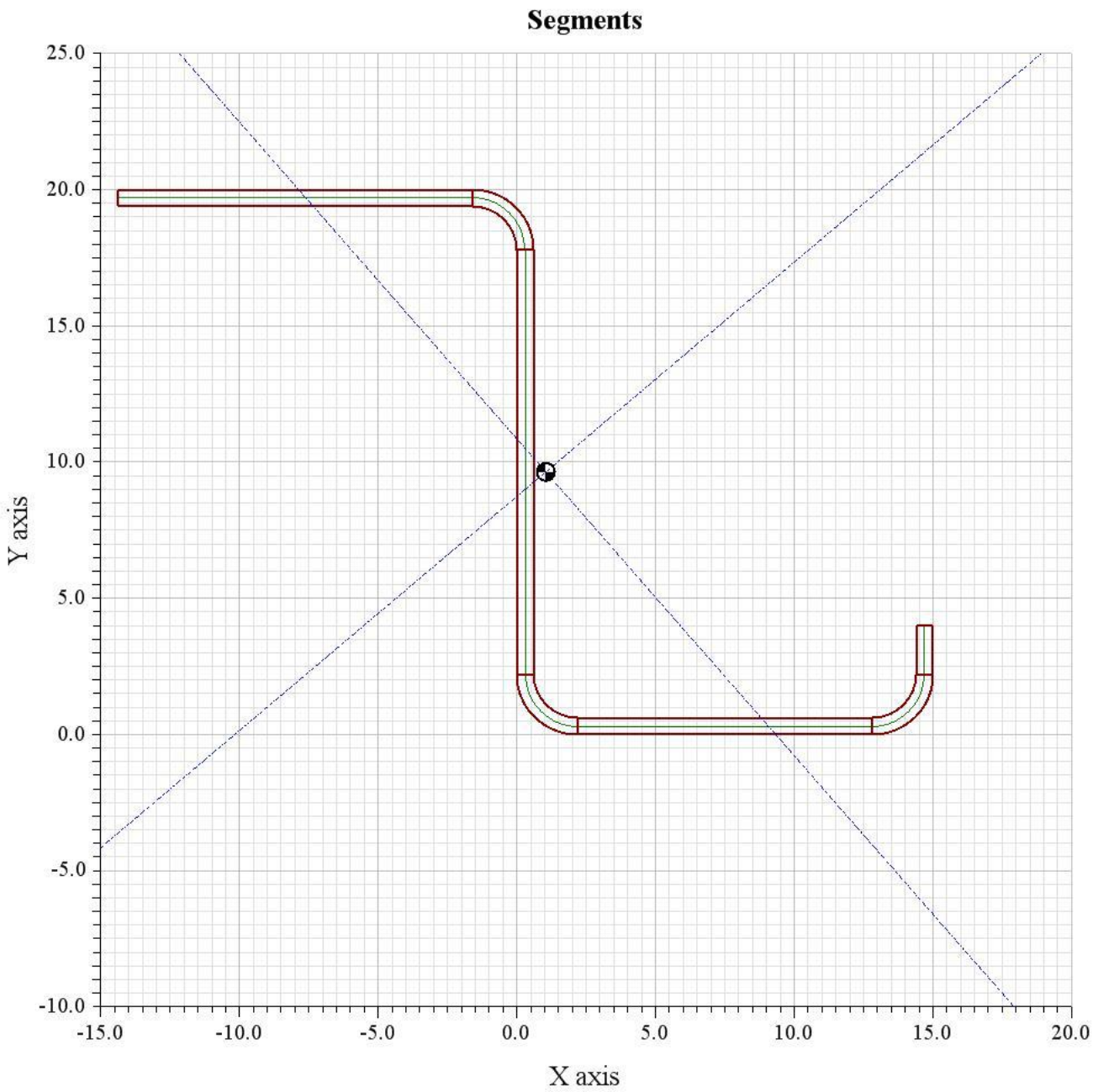
The three buttons located at the lower right corner are to write the results in a file (left button), to draw the section in a graphic file (mid button) or to write the results in a printer (right button).

7.- Examples

In the following examples, the geometric properties of four different sections are calculated. The sections of the first and the second examples are defined by segments and the other sections are defined by contour lines. These examples clarify the entire procedure explained in this Manual.

7.1 Example 1

This section has been defined by segments and it has the following geometry:



The input data file as well as the output results file are provided on next pages:

Input file:

```
1  Section defined by segments: N / Xi   Yi   Xf   Yf   R   t
   7  -14.4   19.7   -1.6   19.7   0.0   0.6
       -1.6   19.7    0.3   17.8  -1.9   0.6
        0.3   17.8    0.3    2.2   0.0   0.6
        0.3    2.2    2.2    0.3   1.9   0.6
        2.2    0.3   12.8    0.3   0.0   0.6
       12.8    0.3   14.7    2.2   1.9   0.6
       14.7    2.2   14.7    4.0   0.0   0.6
```

Output file:

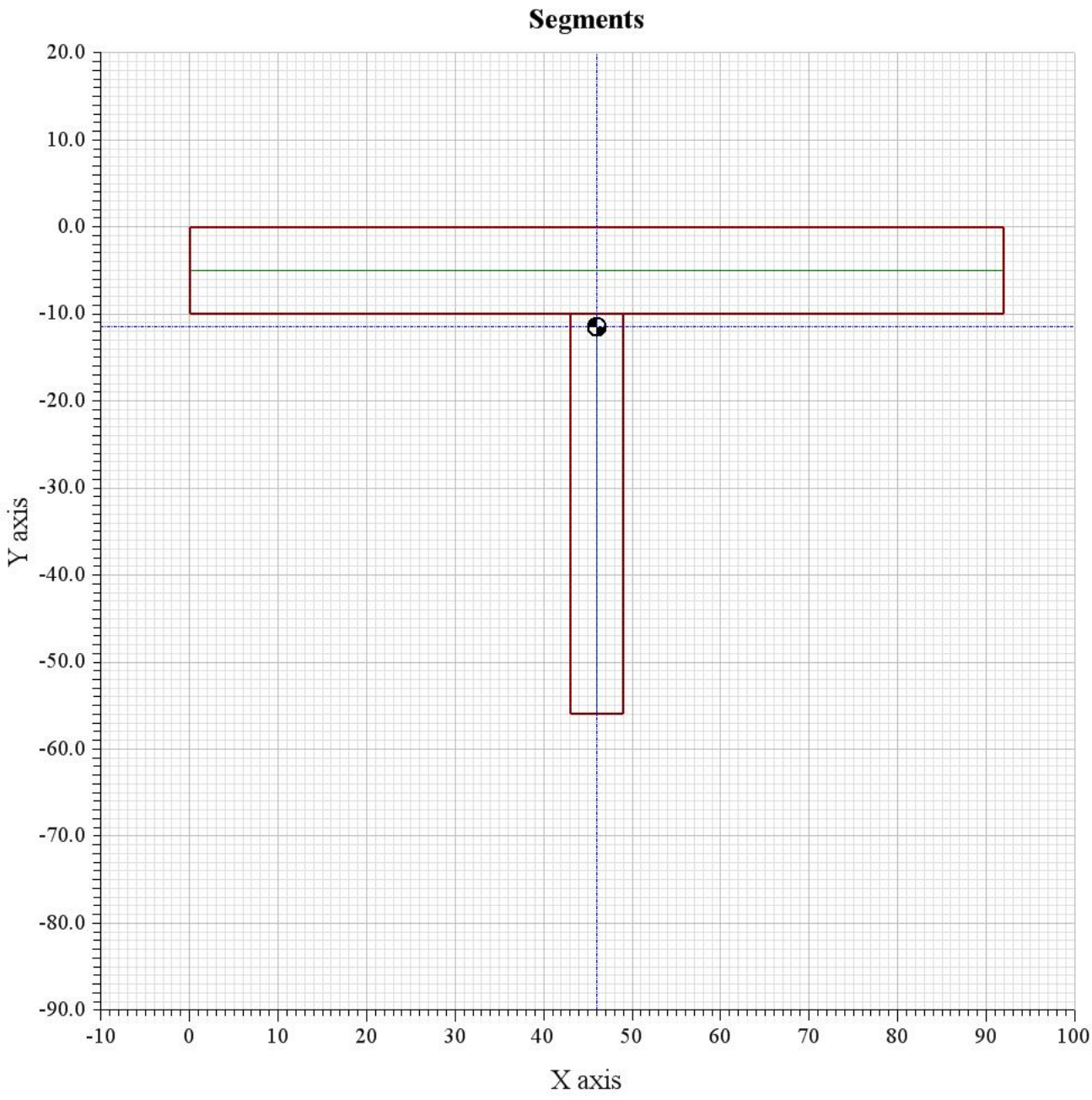
```
**********************************
*                                *
*   <<< Section definition >>>   *
*                                *
**********************************

Section defined by segments.
```

Seg	Xi	Yi	Xf	Yf	R	t
1	-14.400	19.700	-1.600	19.700	0.0000	0.6000
2	-1.600	19.700	0.300	17.800	-1.9000	0.6000
3	0.300	17.800	0.300	2.200	0.0000	0.6000
4	0.300	2.200	2.200	0.300	1.9000	0.6000
5	2.200	0.300	12.800	0.300	0.0000	0.6000
6	12.800	0.300	14.700	2.200	1.9000	0.6000
7	14.700	2.200	14.700	4.000	0.0000	0.6000

```
Units: mm


**********************************
*                                *
*   <<< Section properties >>>   *
*                                *
**********************************

Area of the section:
--------------------
A =  29.852 mm2

Center of gravity position:
---------------------------

XCG =   1.043 mm
YCG =   9.638 mm

Moments of inertia in the center of gravity:
--------------------------------------------

IXX =  1998.0 mm4
IYY =  1571.6 mm4
PXY = -1405.9 mm4

Moments of inertia in principal axes:
-------------------------------------

I1 =  3206.8 mm4
I2 =   362.9 mm4

ANG =  40.69 °
```

7.2 Example 2

This section has been defined by segments and it has the following geometry:



The input data file as well as the output results file are provided on next pages:

Input file:

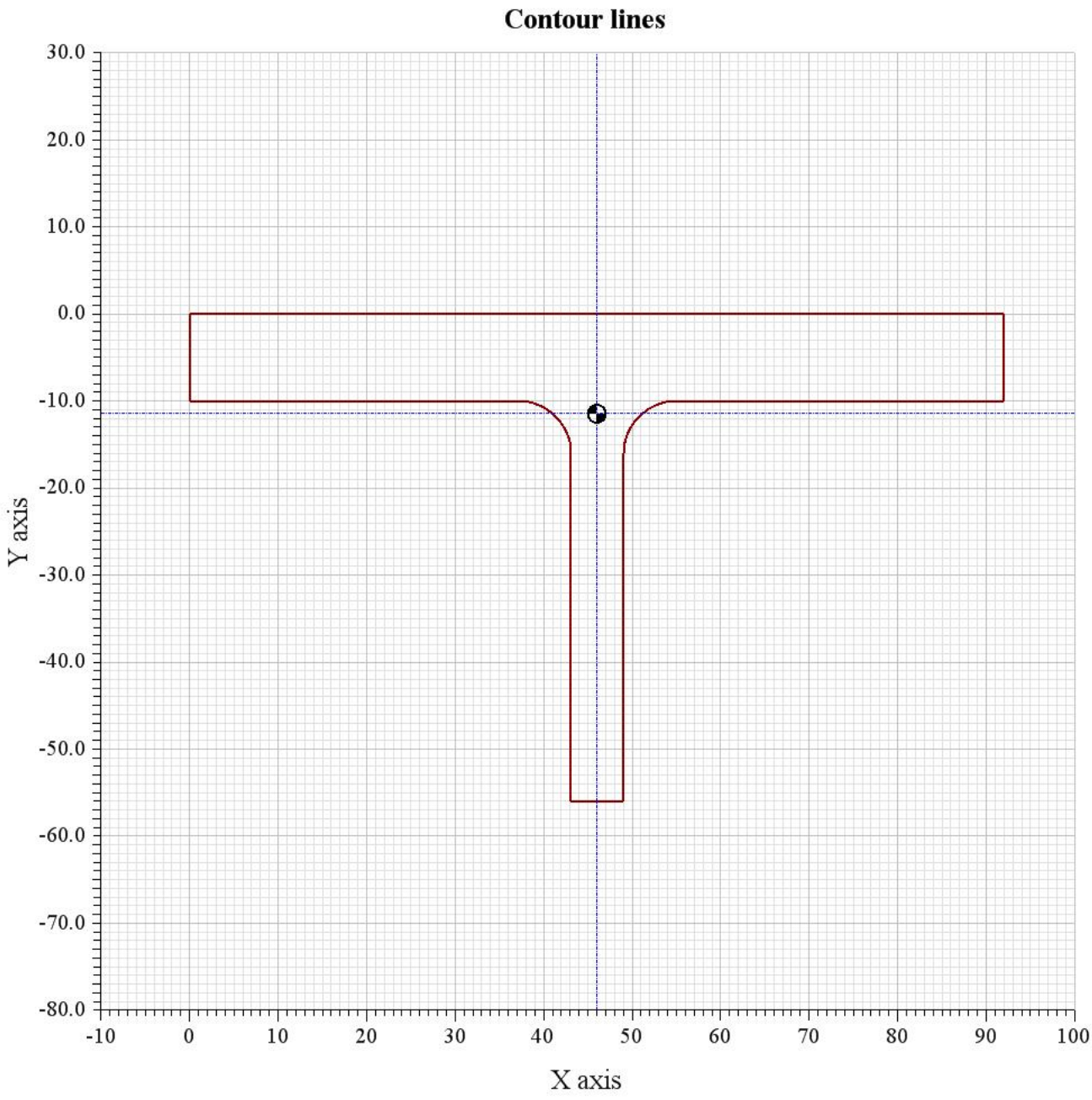
```
1  Section defined by segments: N / Xi   Yi   Xf   Yf   R   t
   2     0.0    -5.0    92.0    -5.0  0.0    10.0
        46.0   -10.0    46.0   -56.0  0.0     6.0
```

Output file:

```
**********************************
*                                *
*   <<< Section definition >>>   *
*                                *
**********************************

Section defined by segments.

 Seg        Xi            Yi            Xf            Yf            R             t
-----   -----------   -----------   -----------   -----------   -----------   -----------
   1         0.000        -5.000        92.000        -5.000          0.00        10.000
   2        46.000       -10.000        46.000       -56.000          0.00         6.000

Units: mm


**********************************
*                                *
*   <<< Section properties >>>   *
*                                *
**********************************

Area of the section:
--------------------
A =  1196.0 mm2

Center of gravity position:
---------------------------

XCG =    46.0 mm
YCG =   -11.5 mm

Moments of inertia in the center of gravity:
--------------------------------------------

IXX =  222783.9 mm4
IYY =  649734.7 mm4
PXY =       0.0 mm4

Moments of inertia in principal axes:
-------------------------------------

I1 =  222783.9 mm4
I2 =  649734.7 mm4

ANG =   0.00 °
```

7.3 Example 3

This section has been defined by segments and it has the following geometry:



The input data file as well as the output results file are provided on next page3:

Input file:

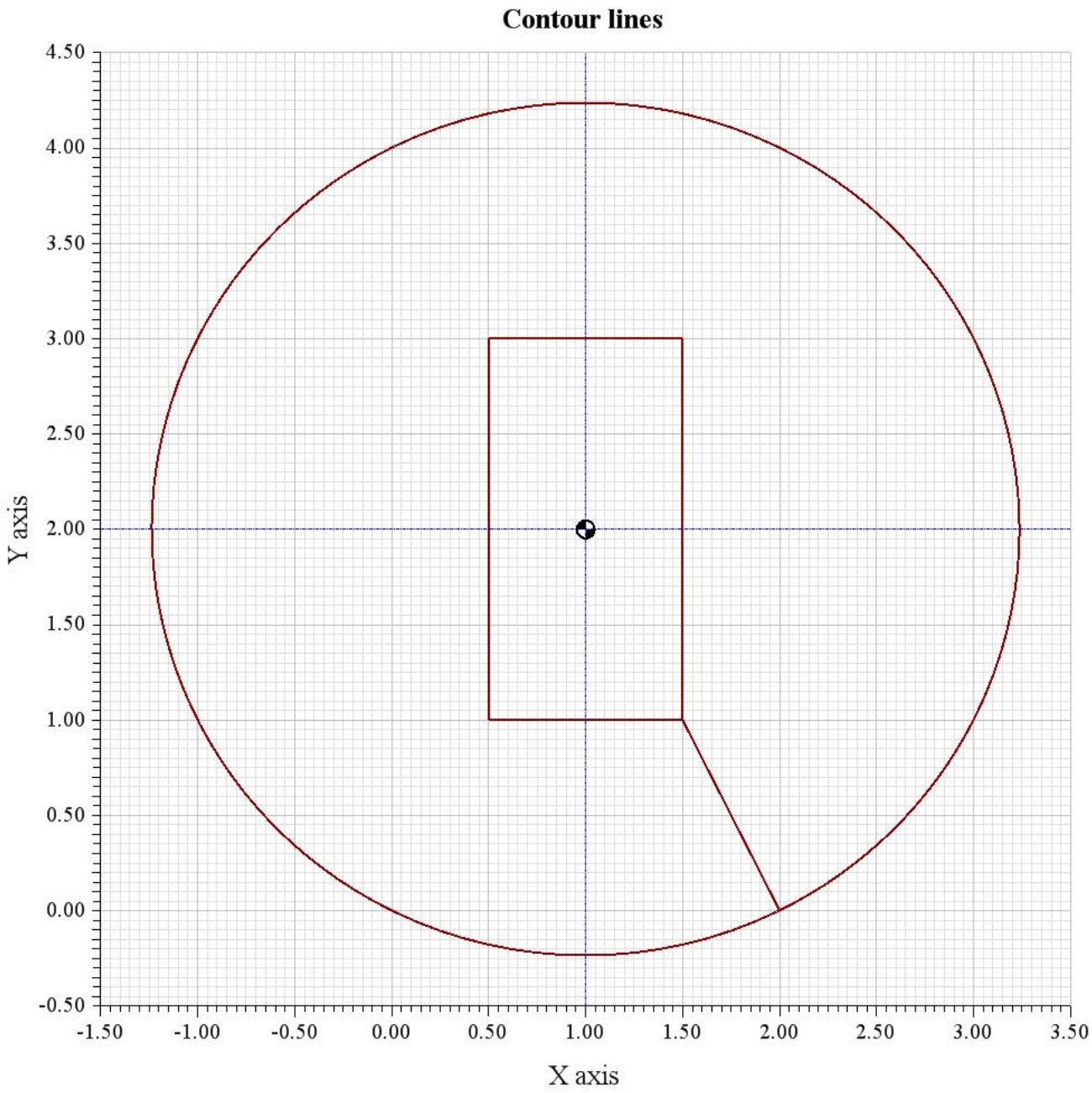
```
2  Section defined by contour lines: N / X  Y  R
   10    0.0    0.0   0.0
        92.0    0.0   0.0
        92.0  -10.0   0.0
        55.0  -10.0   6.0
        49.0  -16.0   0.0
        49.0  -56.0   0.0
        43.0  -56.0   0.0
        43.0  -16.0   6.0
        37.0  -10.0   0.0
         0.0  -10.0   0.0
```

Output file:

```
**********************************
*                                *
*   <<< Section definition >>>   *
*                                *
**********************************

Section defined by contour lines.

Point        X             Y             R
-----   -----------   -----------   -----------
   1         0.000         0.000        0.0000
   2        92.000         0.000        0.0000
   3        92.000       -10.000        0.0000
   4        55.000       -10.000        6.0000
   5        49.000       -16.000        0.0000
   6        49.000       -56.000        0.0000
   7        43.000       -56.000        0.0000
   8        43.000       -16.000        6.0000
   9        37.000       -10.000        0.0000
  10         0.000       -10.000        0.0000

Units: mm


**********************************
*                                *
*   <<< Section properties >>>   *
*                                *
**********************************

Area of the section:
--------------------
A =  1211.5 mm2

Center of gravity position:
---------------------------

XCG =    46.0 mm
YCG =   -11.5 mm

Moments of inertia in the center of gravity:
--------------------------------------------

IXX =  222803.7 mm4
IYY =  650045.1 mm4
PXY =       0.0 mm4

Moments of inertia in principal axes:
-------------------------------------

I1 =  222803.7 mm4
I2 =  650045.1 mm4

ANG =   0.00 °
```

7.4 Example 4

This section has been defined by segments and it has the following geometry:



The input data file as well as the output results file are provided on next pages:

Input file:

```
2  Section defined by contour lines: N / X  Y  R
   10   0.0   0.0   2.2361
        2.0   0.0   0.0000
        1.5   1.0   0.0000
        0.5   1.0   0.0000
        0.5   3.0   0.0000
        1.5   3.0   0.0000
        1.5   1.0   0.0000
        2.0   0.0   2.2361
        2.0   4.0   2.2361
        0.0   4.0   2.2361
```

Output file:

```
**********************************
*                                *
*   <<< Section definition >>>   *
*                                *
**********************************

Section defined by contour lines.

Point        X             Y             R
-----   -----------   -----------   -----------
   1        0.0000        0.0000        2.2361
   2        2.0000        0.0000        0.0000
   3        1.5000        1.0000        0.0000
   4        0.5000        1.0000        0.0000
   5        0.5000        3.0000        0.0000
   6        1.5000        3.0000        0.0000
   7        1.5000        1.0000        0.0000
   8        2.0000        0.0000        2.2361
   9        2.0000        4.0000        2.2361
  10        0.0000        4.0000        2.2361

Units: mm


**********************************
*                                *
*   <<< Section properties >>>   *
*                                *
**********************************

Area of the section:
--------------------
A =  13.708 mm2

Center of gravity position:
---------------------------

XCG =   1.000 mm
YCG =   2.000 mm

Moments of inertia in the center of gravity:
--------------------------------------------

IXX =  18.969 mm4
IYY =  19.468 mm4
PXY =   0.000 mm4

Moments of inertia in principal axes:
-------------------------------------

I1 =  18.969 mm4
I2 =  19.468 mm4

ANG =   0.00 °
```